

BITS CS + SS HD PYQs

Compiled Question Bank

Unofficial Document & Disclaimer

This document constitutes an unofficial, crowdsourced compilation of prospective examination questions aggregated from public communication channels, including Facebook and WhatsApp groups. It is intended strictly for supplementary educational reference and has not been endorsed, vetted, or officially published by the institute.

The author assumes no liability for the veracity, completeness, or accuracy of the questions or their corresponding solutions. Given the decentralized and informal nature of its sourcing, users must acknowledge that certain items may be fragmented, structurally incomplete, or entirely erroneous. It is strongly advised that this material be utilized with the utmost discretion and independent academic judgment.

1 Core Mathematics

- Given a vector equation, find the torsional equation.
- Find the roots of $f(z) = \sin(z) = 0$.
 - $n\pi$
 - $n\pi i$
 - $2n\pi$
 - $2n\pi i$
- For what value is the even part of the function $f(n) = u[n] - (n - 4)$ guaranteed to be zero?
 - $|n| < 3$
 - $|n| < 4$
 - $|n| > 3$
 - $|n| > 4$
- Solve for z : $\cosh(z) + \sinh(z) = -i$.
- Evaluate i^i . (Answer provided: $e^{-\pi/2}$)
- Find the maxima of $f(z) = 6z^{1/3} - 12z^{4/3}$ in the interval $[-1, 1]$.
- Identify the equation given:
 - Circle with center $(1, 1)$, radius 1
 - Ellipse
 - Satisfies $(x, y) = (0, 0)$
 - Straight line
- A conic equation is given in the form $r = \frac{25}{1 + \cos \theta}$. Find its directrix and eccentricity.
- Find the area of the region inside the unit circle but outside the curve $r = 1 - \cos \theta$.

10. Calculate the length of an arc in polar coordinates.
11. At what value does the series $\sum \frac{1}{n^p}$ converge or diverge?
12. Find the residue of $z \cdot \cos(1/z)$.
13. Find the area inside and outside of a circular region, and the area within a closed curve.
14. Find the base x in the following equality equation: $(211)_x = (345)_8$.
15. Identify whether a given function is odd or even.
16. Calculate the residues and transformations for given complex variables.
17. Solve a given Cauchy-Euler equation.
18. Evaluate a complex integration and partial derivatives.
19. Apply the Mean Value Theorem (MVT) and Green's theorem to find the required z value and span.
20. Questions evaluating eigenvectors, continuous distributions, matrix rank, and vector spaces in Linear Algebra.
21. Find the Laplace transform, and solve cylindrical symmetry and helix-based questions.

2 English Language and Logical Reasoning

1. Rearrange 5 jumbled phrases to form a coherent sentence.
2. Fill in the blank: "What job will you apply ___ in the future?"
 - A) for
 - B) to
 - C) among
3. Draw a logical conclusion from a given passage.
4. Select the sentence from the options that has a similar meaning to the given sentence.
5. Correctly use prepositions in phrases like "died of/from some disease" and "apply for/to work".
6. Identify the grammatical difference between "few" and "a few".
7. Determine whether theories are "accepted" or "refuted".
8. Identify the correct meaning of the word "circumspect".
9. Find the errors in three given sentences.
10. Supply the correct connecting words for a given paragraph.
11. Logical Reasoning questions involving completing a figure, numerical patterns, spelling, and series completion.
12. Identify the synonym for a given word from four options.

3 Discrete Mathematics

1. In a bag of 3 white and 5 red balls, two are picked up. For every white ball picked up, you get Rs. 10, and for a black (red) ball, Rs. 35. Find the expected value.
2. In a Poisson distribution, if $P(x = 1) = P(x = 2)$, find $P(x = 4)$. (Answer provided: $\frac{2}{3}e^2$)
3. In a council, the probabilities of election winning for A, B, and C are $\frac{1}{3}$, $\frac{2}{9}$, and $\frac{4}{9}$, respectively. The chances of ladies getting admitted to the council are 0.2, 0.3, and 0.5 if A, B, or C win, respectively. What is the probability of women getting admitted?
4. Given $P(A')$, $P(B')$, and $P(A \cup B)$, find $P(A \cap B)$.
5. What is the probability that three people share the same birthday (i.e., same day, same month) assuming 365 days in a year?
 - A) $1/365$
 - B) $1/(365)^2$
 - C) $1/(365)^3$
6. Find the number of circular arrangements for 8 people if 2 of them always sit together. (Answer provided: $6!2!$)
7. In how many ways can 12 apples be distributed among 4 children such that every child gets at least 2 apples?
8. For a continuous probability density function $p(x) = kx^2$, find the expected value $E(x)$.
9. A table of $p(X)$ is given with values $p(0), p(1), p(2), p(3)$ equal to $k \cdot C$, where k is a constant. Find $P(X < 2)$.
10. How many different 8-bit binary strings contain exactly 5 ones?
11. Find the number of decision trees that can be formed from n boolean variables. (Answer provided: 2^{2^n})
12. From a collection of 20 oranges, 18 apples, and 3 pineapples, in how many ways can you choose a dozen fruits?
13. Find the number of possible relations in $A \times B$ given $A = \{2, 3\}$ and $B = \{a, b\}$.
14. Let P and Q be two mutually exclusive events; given $P(A)$ and $P(B)$, find $P(A' \cap B')$.
15. There are three foods out of which two are alike. A man always selects the odd one. What is the probability that in 5 trials he will choose 4 or more correct foods?
16. Given two sets $M = \{A, B, C\}$ and $N = \{1, 2, 3\}$, find the number of one-to-one (1:1) functions between them.
17. Calculate the probability based on the number of trials and property of inclusion and exclusion.
18. Questions using quantifiers from discrete mathematics, mean and variance, and regression coefficients.

4 Theory of Computation

1. Identify if a Context-Free Grammar (CFG) is closed under union, complement, etc.
2. What is the complexity of converting an NFA to a DFA, and a DFA to a regular expression?
3. What elements are present in a DFA state transition diagram?
 - A) Initial state
 - B) Final state
 - C) Strings
 - D) etc.
4. Let M be a CFG. Which of the following is undecidable?
 - A) $L(M)$ is regular
 - B) $L(M)$ is finite
 - C) $L(M)$ is empty
 - D) All of the above
5. Find the number of states in a DFA for the regular expression a^*b^* .
6. Theory question based on the pumping lemma for regular languages.
7. Questions identifying Context-Free Languages (CFL) and Regular Languages (RE).
8. Questions on Deterministic Pushdown Automata (DPDA) and Non-deterministic Pushdown Automata (NPDA).

5 Programming in C

1. Predict the output of code involving pointers, structures, unions, linked lists, stacks, nested loops, switch cases, and macros.
2. Which storage class cannot use the `&` operator?
 - A) static
 - B) register
 - C) auto
3. Which storage class maintains the value between function calls?
 - A) `int i = something`
 - B) `static int i = something`
4. What is the difference between `fflush()` and `fseek()` for random file rendering?
5. How is the `fgets()` function used in C?
6. Which is not a standard C function?
 - A) `ftell()`
 - B) `fforward()`
 - C) `frewind()`

D) `fopen()`

7. What is the size of the following union, assuming `int` is 4 bytes, `char` is 2 bytes, and `double` is 5 bytes?

```
union s {
    int a;
    char c;
    double d;
};
```

8. What will be the value of `x` given the following macros:

```
#define f(A,B) A+B
#define g(A,B) f(A,B)/f(A,B)
int x = 5;
h(x, x++);
```

9. Predict the output of the following pointer swap code:

```
void swap(int *a, int *b) {
    int *t;
    t = *a;
    *a = *b;
    *b = t;
}

void run() {
    int x = 10, y = 20;
    swap(&x, &y); // Assumed corrected syntax based on typical problem
    printf("%d %d", x, y);
}
```

10. Given a `fork()` question: `int x = 5; fork(); if (pid != 0) print x; else if (pid == 0) x++;` (Answer provided: 5).

6 Data Structures

- What is the maximum difference between the heights of two nodes of an AVL tree?
A) 1
B) 2
C) 3
D) None of the above
- If you have a stream of data and you want to reverse it, which ADT will you use?
A) Stack
B) Queue
- Given the inorder traversal (ABCDEFGH) and postorder traversal (ACBEGFD) of a tree, find the preorder traversal.
- Which traversal sequence gives us elements in sorted order?

5. Identify the overflow and underflow conditions for a circular queue.
6. From a single array, two queues are implemented (e.g., $front1 = -1$ and $front2 = size$). What is the overflow condition?
7. You have Stack 1 growing from the right and Stack 2 growing from the left. Find the condition for overflow.
8. What is the searching time complexity in a linked list?
 - A) $O(n)$
 - B) $O(1)$
 - C) $O(n^2)$
9. Items are removed from a stack, inserted into a queue, dequeued, and pushed again to the stack. What element is at the top of the stack?
10. Which of the following is false about the array representation of a stack?
 - A) push and pop are constant time operations
 - B) PUSH requires checking array out of bounds
 - C) POP requires checking array out of bounds
11. Hashing with linear chaining uses the hash function $h = k\%9$. Find the maximum length of the linked list formed. (Answer provided: 4)
12. Calculate the minimum number of internal nodes in an AVL tree with height 5.
13. Find the height of a binary tree with n nodes where each internal node has exactly 2 children.
14. Which sequence, if inserted in an AVL tree, will cause no adjustment in the tree?
 - A) 1 2 3 4 5
 - B) 5 4 3 2 1
 - C) 3 2 4 1 5
 - D) 4 3 1 2 5
15. What is the difference between an Array and a Linked List?
16. Identify the properties of a full binary tree and a complete binary tree.
17. Questions on 3-dimensional arrays and calculating the address of a location.

7 Algorithms

1. Which of the following cannot be used as a heuristic in optimization?
 - A) Number of states
 - B) Maximum depth in state space tree
 - C) Height of root
 - D) Largest height
2. What is the time complexity of finding the distance between two vertices using the adjacency matrix representation of a graph?

- A) $O(n)$
 - B) $O(n^2)$
 - C) $O(\log n)$
 - D) $O(1)$
3. What is the time complexity of DFS on a complete graph using an adjacency list? (Answer provided: $O(V^2)$)
 4. Find the time complexity of the function $T(n) = 2T(n^{1/2}) + \log n$.
 5. Find the number of paths for the Traveling Salesman Problem with 10 vertices.
 6. Given two complexity functions $f(n) = n^2 - n + 550$ and $g(n) = 59n + 50$, for what values of n does $f(n)$ run in less time compared to $g(n)$?
 7. To sort the following numbers, which algorithm will suit the best: (i) 1 to 100 integers, (ii) 0 to 1,000,000 integers?
 - A) Bucket sort for both
 - B) (i) Radix sort (ii) Quick sort
 - C) (i) Quick sort (ii) Merge sort
 - D) (i) Merge sort (ii) Quick sort
 8. Find the time complexity to find the median of 10 sorted elements in an array.
 9. Which of the following is false?
 - A) For every recursive algorithm, there is an iterative algorithm.
 - B) For every iterative algorithm, there is a recursive algorithm.
 - C) A recursive solution uses less space than an iterative solution.
 10. Determine what is meant by a stable sorting algorithm.

8 Digital Electronics and Microprocessors

1. If we reverse both the input and output of an XOR gate, which gate will it become?
2. What are the three output states of logic gates?
 - A) High
 - B) Low
 - C) High impedance
 - D) Low impedance
3. Which of the 4 given expressions boils down to the Product of Sums (POS) form?
4. Which memory is used to increase the CPU speed, reduce I/O operations, or increase memory capacity?
 - A) Cache
 - B) RAM
5. What is the instruction to put the 8085 microprocessor in sleep mode?

- A) HALT
 - B) HLD
 - C) NOP
6. In 8085, which instruction is used to deactivate the microprocessor until an interrupt is received?
- A) HLT
 - B) HOLD
7. What does the HOLD signal do in an 8086 microprocessor?
8. What does the LOCK signal mean in an 8086 microprocessor?
9. When the 8259 works in cascaded mode with 8086, what is required?
- A) ICW1 & ICW2 are required
 - B) ICW1, ICW2, ICW3 & ICW4 are required
 - C) ICW1 & ICW2 are required
 - D) ICW1, ICW2 & ICW3 are required
10. Determine the difference between a latch and a flip-flop (voltage difference properties).
11. Find the size of the ROM required for converting binary code to Gray code.
- A) 8×4 bits
 - B) etc.
12. Given the Gray code of an unknown binary code, find the Excess-3 code for that binary code.
13. Find the minterm of 15 and maxterm of 12 for the function $f(A, B, C, D)$.
14. An explanation about a mystery digital logic device was given and asked to identify it.
(Answer provided: Multiplexer)

9 Operating Systems

1. The Operating System is transparent to what?
- A) RAM
 - B) Cache
 - C) CPU
 - D) None
2. Which of the following transitions is not valid?
- A) Ready to Running
 - B) Blocked to Running
 - C) Running to Blocked
3. Two processes, P1 and P2, are running according to Shortest Remaining Time First (SRTF). If P1 wants to do an I/O operation, which of the following is correct regarding P2?

4. Find the number of page faults using the FIFO and LRU page replacement policies for a given string.
5. Calculate the external fragmentation for a given page size of 4KB.
 - A) 0
 - B) 10
 - C) etc.
6. What is a zombie process?
7. Which of the following statements is true about threads in an OS?
 - A) It is a lightweight process.
 - B) Multiple processes can share the same thread.
 - C) One process can have at most one thread.
8. Under which condition will a deadlock not occur?
9. Which of the following is deadlock-free?
 - A) 2PL
 - B) Wait and Die
 - C) Strict 2PL
10. You are given two `for` loops manipulating a shared variable $X = 0$. Find the minimum and maximum values of X after the following concurrent code is executed:


```
For i=0 to 10 => X++; X--
For j=0 to 10 => X++; X--
```
11. Given the burst time of 5 processes arriving at $t = 0$, find the average waiting time using SJF scheduling.
12. Given the burst time of 5 processes arriving at $t = 0$, find the average Turnaround Time using Round Robin scheduling.
13. What is the wait-for graph used for? (Answer provided: Deadlock detection)
14. What is a strong semaphore?
15. What happens during a context switch (what kinds of activities occur)?
16. Questions on the critical section problem, asynchronous I/O, paging (TLB, Level 1 & Level 2), and number of page tables required.

10 Computer Organization and Architecture

1. What is the ideal Cycles Per Instruction (CPI) of a 5-stage RISC pipeline? (Answer provided: 1)
2. Which of the following is true regarding pipelines? (Answer provided: Low CPI and high clock rate are preferred)
3. What factors is pipelining performance based on?
4. In a matrix multiplying operation:

- A) Code uses temporal locality, data uses spatial locality
 - B) Data uses temporal locality, code uses spatial locality
 - C) Code and data both use temporal locality
 - D) Both use spatial locality
5. When adding two negative numbers, if a carry is generated, what does it say about the result and the overflow bit?
 6. Which of the following is not a basic part of the von Neumann architecture?
 - A) Network card
 - B) Input device
 - C) CPU
 - D) Output device
 7. Given a 4-way set associative cache, memory address size, block size, and main memory size, calculate the number of tag bits.
 8. A hard disk of capacity 30GB contains 4096 tracks per platter, 2048 sectors per track, and 1024 bytes per sector. How many platters will be required?
 9. During which phase of the instruction cycle does the Program Counter (PC) change?
 10. Read and write signals are part of which bus?
 - A) Control
 - B) Data
 - C) Address
 11. For a computer address, if we reduce the number of bits by 2 and double the addressability, then what will happen to the total address space?
 12. If you have a 256-terabyte virtual address space, how many bits are used to represent it?
 - A) 32
 - B) 48
 - C) 47
 - D) 16
 13. What is the largest range for a 15-bit 2's complement number?
 - A) 2^{15}
 - B) $2^{15} - 1$
 - C) etc.
 14. Difference between striping and mirroring in RAID. Which statement is incorrect?
 15. Questions related to Big-endian vs Little-endian combined with unions.

11 Computer Networks

1. Which cable CANNOT be used for computers?
 - A) Cat1
 - B) Cat7
 - C) Cat5
 - D) Cat5e
2. What is the port number used by the SQL server?
3. What is the maximum size of a UDP packet (Header + Data)?
 - A) > 65535
 - B) No upper limit
 - C) < 65535
4. Which protocol is used to map an IP address to a MAC address?
 - A) ARP
 - B) RARP
 - C) SMTP
5. At which OSI layer does Ethernet 802.3 operate?
6. Which port cannot be used by a client for HTTP?
 - A) Random values
 - B) 80
7. Which is the correct statement about HTTP?
 - A) HTTP uses port 8080 for listening.
 - B) HTTP uses port 80 for sending.
8. Find the data downloaded given a bandwidth of 56kbps and a time of 2 minutes.
9. Given an IP address and a subnet mask (CIDR block), find the subnet ID and broadcast address.
10. Calculate the efficiency of Go-Back-N given bandwidth, RTT, frame length, and window size.
11. Which of the following does not affect LAN cable length?
 - A) Signal propagation speed
 - B) Slot time
12. If the webpage `www.bitspilani.ac.in` has 10 objects and HTTP v1.1 is used, how many TCP connections are established to send the objects?
 - A) 10
 - B) 1
 - C) 0
13. Where is an ICMP message (e.g., Destination Unreachable) enclosed by the router?

A) First in IP then in frame

Other options

14. Questions covering distance vector routing, CSMA/CD, 10Base2 Ethernet properties, and the differences between TCP and UDP.

12 Database Systems

1. Which of the following schemas provides the highest level of abstraction in a database system?
 - A) Internal Schema
 - B) Logical Schema
 - C) External Schema
 - D) Physical Schema
2. In an ER Model, which of the following is correct?
 - A) There can be multiple physical schemas and a single logical schema.
 - B) Single physical schema and multiple logical schemas.
 - C) Both multiple.
3. In an ER diagram with 'Hotel Room' and 'Persons' as entities and 'Lodging' as a many-to-many relationship, where is the attribute 'Total rent to be paid by a guest' most likely stored? (Answer provided: In Lodging)
4. Which of the following is not a valid SQL join?
 - A) Union
 - B) Product
 - C) Left Join
 - D) Right Join
5. Given a relation $R(A, B, C, D)$ where A and B are unique and not null, which statement about functional dependencies is true?
6. If a relation is decomposed into BCNF, what is its level of redundancy?
7. Which of the following do not suffer from cascading rollbacks?
 - A) 2PL
 - B) Strict 2PL
 - C) Basic 2PL
 - D) All of the above
8. If a set of attributes in a relation R gives R itself, what is that set of attributes called?
 - A) Primary Key
 - B) Super Key
 - C) Candidate Key
 - D) Foreign Key
9. Do checkpoints reduce undo work, redo work, both, or none?

10. Compare the number of records in a dense index file versus the indexed file.
 - A) Same
 - B) More than
 - C) Less than
11. A relation has attributes Car Model and Car color. What is the result of the query:
Select colors from R group by colors?
12. Questions on Two-Phase Locking (2PL), shadow paging, wait-die scheme, and ER diagrams.

13 Compiler Construction

1. Which language is used to describe the semantics and syntax of C?
 - A) Attribute grammar
 - B) Syntax tree
 - C) Context-Free Grammar
2. What is the responsibility of a type checker in a compiler?
3. A compiler encounters every character in which phase?
4. What is the purpose of a linker in a compiler?

14 Object Orientation and Software Engineering

1. What is requirement gathering?
2. Which are not the qualities of a good SRS (Software Requirements Specification)?
3. Which of the following is not a testing method?
 - A) Clear box
 - B) Color box
 - C) White box
 - D) Black box
4. What is the walkthrough approach in testing?
5. Identify the correct design pattern (e.g., Factory pattern) for specific use cases.
6. Provide the definition of inheritance, Object, and Class.
7. What approach is COCOMO II based on?
 - A) Top-down
 - B) Bottom-up
 - C) Algorithmic
 - D) Complex
8. Who performs Beta testing?
 - A) Software developers

- B) App developers
 - C) Testers
 - D) Customers
9. The lifecycle of a class is shown by which UML model?
- A) Sequence
 - B) Class
 - C) Component
 - D) Collaboration diagram
10. What are the features of good software design?
- A) High cohesion & low coupling
 - B) Low cohesion & high coupling
 - C) High cohesion & high coupling
11. Which of the following is not a software architecture?
- A) Facade
 - B) Filters & pipes
 - C) Microkernel
12. The Waterfall model is not suitable for:
- A) Simple Projects
 - B) Complex Projects
 - C) Accommodating changes
 - D) Medium Projects
13. When people are involved in the development process, which evolutionary model is used?

15 Topics Outside the Above

1. **Artificial Intelligence:** Which of the following scenarios with respect to an autonomous taxi driving system are examples of unsupervised learning?
 - A) Learning traffic distribution with time observing traffic pattern
 - B) Learning to drive effectively using feedback
 - C) Learning speed control in the presence of an instructor
 - D) Learning taxi fares with distance using examples
2. **Artificial Intelligence:** Find the Markovian blanket of a particular node in a given Bayesian Network.
3. **Artificial Intelligence:** What is 'common sense' in AI? Questions on reactive agents and Constraint Satisfaction Problems (CSP).
4. **Miscellaneous:** What is the definition of XML? How does it differ from HTML?
5. **Miscellaneous:** What is a virtual machine?